Wednesday Lab Assignment

1. Write a C program to input any character and check whether it is a number or a alphabet or neither any of those.
2. Write a program to input 4 numbers and print out the largest and the second largest number.
3. Write a program to input a 3 digit number and reverse it. And check if the number is palindrome or not.
4. Given the age of a person as x and the gender as 'M' or 'F'. Check whether the person is between 18 and 60 yrs. If person is between 18 and 60 then check if the person is male or female and if person is male print 'Male' other wise print 'female'. If person is not between 18 and 60 and the age is less than 18 print 'too small' otherwise print 'too big'.
5. Take 4 numbers as an input and calculate the distance from fourth input to all other three inputs. Find the two nearest inputs to the fourth input among the other three inputs.
6. Write a C program to find the roots of a quadratic equation of the form $ax^2 + bx + c = 0$. The program should take the coefficients a, b, and c as input from the user and then calculate the roots based on the discriminant ($D = b^2 - 4ac$). Handle the following cases:
   1. D > 0: The equation has two distinct real roots.
   2. D = 0: The equation has exactly one real root (a repeated root).
   3. D < 0: The equation has no real roots, but two complex roots.
7. Write a C program to calculate and print the electricity bill of a given customer. The program should capture the customer ID, name, and the number of units consumed by the customer from the keyboard. Based on the unit consumption, the program should calculate the total amount to be paid by the customer according to the following rates:

Units Consumed	Charge per Unit
Upto 199	Rs. 1.20
200 and above but less than 400	Rs. 1.50
400 and above but less than 600	Rs. 1.80
600 and above	Rs. 2.00

If bill exceeds Rs. 400 then a surcharge of 15% will be charged and the minimum bill should be of Rs. 100/-

EXAMPLE :

1001

James

800

Expected Output :

Customer IDNO :1001

Customer Name :James

unit Consumed :800

Amount Charges @Rs. 2.00 per unit : 1600.00

Surchage Amount : 240.00

Net Amount Paid By the Customer : 1840.00